# AAKANKSHA KASHYAP

GenAI Engineer | Applied AI Engineer | AI Backend Engineer

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## PROFESSIONAL SUMMARY

AI Engineer with 3.8+ years of experience spanning enterprise backend systems at Ericsson and applied AI systems development in production and personal projects. Builds agentic AI systems using LangGraph orchestrating stateful multi-node architectures with dynamic planning, conditional routing, and multi-tool execution alongside end-to-end RAG pipelines (embeddings, vector retrieval, grounded LLM responses) using Python, FastAPI, and LangChain. Currently building an enterprise grade voice AI receptionist platform at a stealth startup, serving dental, legal, spa, and automotive industries.

## TECHNICAL SKILLS

**AI & Applied AI:** OpenAI APIs, OpenAI SDK, Retrieval-Augmented Generation (RAG), RAG Architecture, Prompt Engineering, Embeddings, ChromaDB, Neon Postgres, pgvector, Retrieval Pipelines, LangChain, AI Agents, AI Automation, LangGraph

**Programming & Backend:** Python, Java, SQL, FastAPI, REST APIs, HTTP/JSON, Pydantic, Async APIs, Validation, Error Handling

**AI & Developer Tools:** n8n, ElevenLabs SDK, Git, Linux, VS Code, Postman, Node.js, JavaScript, WebRTC, Agile, Claude Code, Codex, Vercel, Azure OpenAI (chat + embeddings, v1 API), Azure PostgreSQL Flexible Server + pgvector, Azure Container Apps, Docker + Azure Container Registry

## PROFESSIONAL EXPERIENCE

### AI Engineer | Stealth Startup

*Feb 2026 - Present*

- Integrated agentic orchestration for appointment scheduling, cancellation, and rescheduling and Retrieval-Augmented Generation (RAG) capabilities into a personalized AI voice receptionist platform for US-based B2B clients. Collaborated with the CTO on agentic orchestration for appointment booking workflows using LangGraph, extending the multi-node architecture applied in independent voice AI agent work.
- Designed and implemented end-to-end document ingestion and semantic retrieval pipelines including text extraction, chunking, embedding generation, vector indexing, and similarity search using Python and FastAPI.
- Developed AI backend services enabling domain-specific knowledge retrieval and grounded LLM responses during real-time voice interactions.
- Grounded LLM responses using retrieval-augmented generation (RAG) and domain-specific business knowledge to improve response relevance and reliability.
- Collaborated with the founding team to serve **15+ clients** across industries including dental clinics, spas, legal, and automotive repair businesses.
- Evaluated retrieval quality, response relevance, and end-to-end conversational performance to enhance user experience.
- Implemented guardrails and security scans to safeguard AI responses and protect against prompt injection and data leakage risks.
- Built an AI-powered UGC ad generation workflow as a learning initiative encouraged by the team; later adopted and extended for company marketing content automation.

***Tech Stack:*** *Python, FastAPI, PostgreSQL (pgvector), OpenAI SDK, LangChain, Vector Databases, RAG, Neon Postgres, Azure, n8n*

### Software Developer | Ericsson

*Dec 2022 - Feb 2026*

- Worked on enterprise telecom applications and large-scale network monitoring systems across two projects: Verizon's GCT Tool and Ericsson's NSRS project for Telefonica Sweden.
- End-to-end full stack feature development and enhancement of network monitoring features using Python/Java and backend frameworks, REST APIs, and integrations supporting large-scale telecom network monitoring systems, network servers, and redirect servers.
- Improved 'Central Map View' application user experience with seamless network monitoring, along with improvements in physical link visualization, spontaneous LSP state updates, and context menu integrations.
- Architected the migration of a legacy, global-scope Vanilla JS codebase into a modern ES Module (ESM) architecture. Designed a custom Webpack production pipeline leveraging Tree Shaking, Terser, and Code Splitting, slashing the application payload size by **80%** and eliminating critical global-scope technical debt.
- Wrote Python automation and scripting for CI/CD pipeline tasks across Jenkins and Fortify security scanning workflows.
- Remediated **93.8%** of high and critical Fortify security vulnerabilities across production applications, significantly hardening system security and ensuring enterprise compliance.
- Led migration of codebase from legacy servers to newer RHEL-8 servers and from CORBA to MQTT protocol, enabling lighter and faster communication.

***Tech Stack:*** *Python, Java, Vanilla JavaScript, Spring Boot, FastAPI, REST APIs, SQL, Jenkins, Fortify, CI/CD, Azure, Webpack, Open Street Map*

### Internship | Dopamine Enterprises

*Jan 2022 - Aug 2022*

- Designed SQL schemas and queries for order/customer data on an online water-ordering platform; tested and debugged REST APIs and wrote unit tests.
- Collaborated on full-stack features, frontend-backend integration, API documentation, and Git-based code reviews.

***Tech Stack:*** *Java, Spring Boot, Python, SQL, JavaScript, REST API.*

## AI ENGINEERING PROJECTS

### Multi-Node Graph-Orchestrated Voice AI Agent

**Live Demo:** [https://final-ai-voice-receptionist.vercel.app/](https://final-ai-voice-receptionist.vercel.app/)

- Built an agentic voice AI receptionist using LangGraph to orchestrate dynamic planning, conditional routing, and multi-tool execution across appointment booking, cancellation, rescheduling, and human escalation workflows.
- Designed a stateful multi-node graph architecture (planner, FAQ, RAG retrieval, and tool nodes) enabling context-aware, dynamic decision-making beyond single-prompt or linear-chain LLM calls.
- Implemented a Retrieval-Augmented Generation pipeline (OpenAI embeddings, Neon PostgreSQL, pgvector) to ground conversational responses in business FAQs, services, pricing, and policies.
- Built mock tool integrations (booking, cancellation, rescheduling, human escalation) to simulate realistic backend business operations within the agent's tool-calling layer.

- Engineered end-to-end voice interaction using browser-based Speech Recognition and Speech Synthesis APIs, integrated with a FastAPI backend for real-time request handling.
- Implemented conversation state management to maintain context across multi-turn voice interactions, allowing the agent to track intent and progress through in-flight tasks.
- Architected and deployed the system with a production-inspired folder structure, independent Git history, and its own Vercel deployment as a standalone portfolio project.

***Tech Stack:*** *FastAPI, LangGraph, LangChain, OpenAI, Neon Postgres, pgvector, HTML/CSS/JavaScript, Vercel*

### Enterprise AI Knowledge Assistant

**Live Demo:** [https://enterprise-ai-knowledge-assistant-l.vercel.app/project-rag.html](https://enterprise-ai-knowledge-assistant-l.vercel.app/project-rag.html)

- Built an end-to-end Retrieval-Augmented Generation (RAG) system for answering questions from company knowledge documents including PDFs, policies, SOPs, and onboarding guides.
- Designed and implemented a document ingestion and semantic retrieval pipeline including text extraction, chunking, embedding generation, vector indexing, and similarity search using OpenAI embeddings and PostgreSQL (pgvector).
- Developed REST APIs using FastAPI for document upload, ingestion, retrieval, and conversational question answering.
- Implemented source citations and retrieval-based context injection to improve answer transparency and accuracy.
- Re-implemented the retrieval pipeline using LangChain (document loaders, retrieval chains, prompt orchestration) alongside the original framework-free version, to demonstrate both approaches.

***Tech Stack:*** *Python, FastAPI, PostgreSQL (pgvector), OpenAI SDK, LangChain, Vector Databases, RAG, Neon Postgres*

### Image-to-UGC Ad Generation Workflow

*Originally developed as a learning project; later adopted and enhanced for internal company use.*

**Live Demo:** [https://aakanksha98.github.io/project-ugc.html](https://aakanksha98.github.io/project-ugc.html)

- Built an AI-powered workflow automation system for generating UGC-style video advertisements from a single source image.
- Designed automated AI workflow pipelines for content generation, orchestration, and media processing.

***Tech Stack:*** *n8n, AI Workflow Automation Pipelines, Prompt Engineering*

## EDUCATION

Certified Java Full Stack Developer, Simplilearn (2022)

B.Tech in Information Technology, Techno India College of Technology, Kolkata (2017 - 2021) - CGPA: 8.0

Higher Secondary, Shree Krishna Public School, CBSE (2014 - 2016) - 76.2%

Secondary School, Mount Assisi School, Bhagalpur, ICSE (2002 - 2014) - 92.6%

## ACHIEVEMENTS

- Received "Going Above and Beyond Award" - Stealth Startup recognized for consistently exceeding expectations and driving initiative in a short span of time.
- District Level Topper in National Talent Search Exam (NTSE).